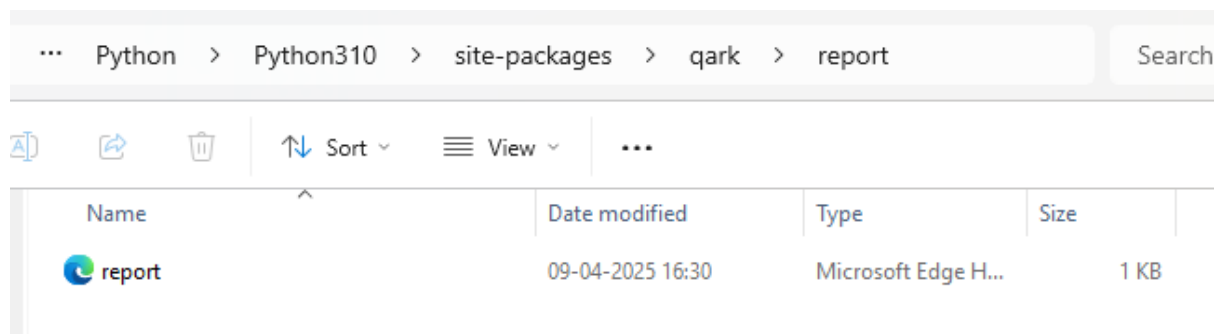


Mobile Security

Qark

```
C:\Users\Admin>qark --apk C:\Users\Admin\Downloads\qark
Decompiling...
Running scans...
Finish scans...
Writing report...
Finish writing report to C:\Users\Admin\AppData\Roaming\Python\Python310\site-packages\qark\report\report.html ...
```

Here we are decompiling the apk file of Diva Application, Decompiling the apk file can help for Static Analysis of the application.



TCPDUMP

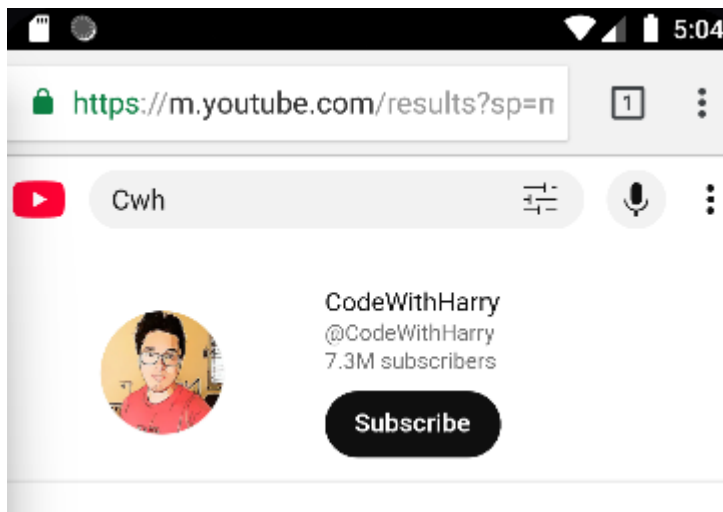
Here we are performing tcpdump inside the adb shell, so we need our system to be rooted.

```
C:\Users\Admin>adb root
adb is already running as root

C:\Users\Admin>adb remount
remount succeeded
```

Let's turn on tcpdump running on the shell and try searching something on the google chrome of the emulator.

```
C:\Users\Admin>adb shell
generic_x86:/ # tcpdump -D
/system/bin/sh: tcpdump: not found
127|generic_x86:/ # tcpdump -D
1.radio0 [Up, Running]
2.wlan0 [Up, Running]
3.any (Pseudo-device that captures on all interfaces) [Up, Running]
4.lo [Up, Running, Loopback]
5.nflog (Linux netfilter log (NFLOG) interface)
6.nfqueue (Linux netfilter queue (NFQUEUE) interface)
generic_x86:/ #
```



And yh we saved the pcap file as dump.pcap

```
generic_x86:/ # tcpdump -vv -i any -s 0 -w /sdcard/dump.pcap
tcpdump: listening on any, link-type LINUX_SLL (Linux cooked), capture size 262144 bytes
^C4880 packets captured
4882 packets received by filter
0 packets dropped by kernel
generic_x86:/ # |
```

Let's pull the pcap file to the Main system for wireshark analysis.

```
^C4880 packets captured
4882 packets received by filter
0 packets dropped by kernel
generic_x86:/ # exit

C:\Users\Admin>adb pull /sdcard/dump.pcap C:\Users\Admin\Downloads\dump.pcap
/sdcard/dump.pcap: 1 file pulled, 0 skipped. 172.5 MB/s (3540948 bytes in 0.020s)

C:\Users\Admin>|
```

Wireshark analysis result :

2031	22.004590	192.168.232.2	142.250.182.238	TLSv1.2	238 Application Data
2032	22.004858	142.250.182.238	192.168.232.2	TCP	56 443 → 39841 [ACK] Seq=691508 Ack=5285 Win=8760 Len=0
2033	22.005908	192.168.232.2	10.0.2.3	DNS	77 Standard query 0x2d27 A www.youtube.com
2034	22.007160	192.168.232.2	10.0.2.3	DNS	77 Standard query 0x19fb AAAA www.youtube.com
2035	22.009525	64.233.170.84	192.168.232.2	TCP	60 443 → 34266 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=
2036	22.009726	192.168.232.2	64.233.170.84	TCP	56 34266 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
2037	22.010029	192.168.232.2	64.233.170.84	TLSv1.2	270 Client Hello (SNI=accounts.google.com)
2038	22.010259	64.233.170.84	192.168.232.2	TCP	56 443 → 34266 [ACK] Seq=1 Ack=215 Win=8760 Len=0